

Artificial Intelligence (AI) is a multidisciplinary field of computer science focused on creating systems capable of performing tasks that typically require human intelligence.

Modern advancements in Natural Language Processing (NLP) and Large Language Models (LLMs) have transformed the way we interact with technology and process information.

Machine learning algorithms build mathematical models based on sample training data to make predictions or decisions.

1. Supervised Learning: Learning a function that maps an input to an output based on example input-output pairs. Common methods include linear regression, logistic regression, and support vector machines.
2. Unsupervised Learning: Discovering hidden patterns or data groupings without labeled responses. Common methods include clustering, principal component analysis, and association rule learning.
3. Reinforcement Learning: Training agents to take actions in an environment to maximize cumulative reward through trial and error.

Deep Learning architectures, particularly multi-layer neural networks and transformers, serve as the foundational building blocks for many modern machine learning applications.

Retrieval-Augmented Generation (RAG) addresses key limitations of static large language models, including hallucinations and outdated information.

The standard RAG pipeline operates as follows:

First, source documents are ingested and split into manageable text chunks. Next, an embedding model transforms these chunks into high-dimensional vectors.

When a student poses a query, the query is embedded into the same vector space, and similarity search identifies the most relevant source documents.